Grey wolf optimizer for allocation and sizing of distributed renewable generation

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***Abstract*—Increasing penetration of distributed energy resources (DERs) have brought operational and control philosophy changes in Smart Grids (SGs). Renewable energy based technologies are becoming more important due to their economic and environmental impacts. Distributed generations (DGs) in the form of small renewable energy resources such as solar photovoltaics (PVs) and Wind Turbines (WTs) are connected in radial distribution networks near to the loads. This paper presents optimal siting and sizing of distributed renewable energy resource to maintain voltage magnitude profiles. Bus voltage magnitude differences for each hour in a day of a distribution system are formulated as an objective function. Three consecutive days are taken into account representing the three seasons of a year. A new nature inspired algorithm Grey Wolf Optimizer (GWO) is used as a solution tool. The proposed formulation is applied to 33 bus and 69 bus radial distribution networks. MATLAB simulations are performed to validate the performance of the approach. Simulation results are discussed and compared with of the several available ones'.**

Keywords— Smart Grids; distributed energy resources (DERs); nature inspired optimization; voltage magnitude problem

I. INTRODUCTION

Penetration of distributed generation (DG) technologies have been increasing day by day because of their potential technical and economic benefits. Among them, avoiding the cost and the loss of power transmission as being located close to the load centers can be assigned as the most remarkable one when compared with large centralized generation stations [1]. DGs also provide environmental-friendly solutions compared to traditional thermal generation due to their less amount of carbon dioxide emission. However, having troubles for getting a license for photovoltaic (PV) units and wind park turbines (WTs) with power output more than 1 MW and weak capability in dispatching and management of DGs are the major challenges for distribution system operators in Turkey.

Several DG types are used in power distribution systems; such as, PV generation, WTs, biomass, fuel cells, geothermal etc. The numerous technical, economic and environmental benefits offered by different renewable DG technologies can be found in [2]. However, high-level electricity penetration from solar and wind in a grid is a great challenge because of some technical, economic and regulatory aspects. Moreover, the location and the size of the units have significant impacts on the expected benefits of DG penetration.

Optimum DG placement and sizing in distribution networks has attracted the researchers during the last decade. The problem was formulated either as a single or a multi-objective optimization to optimize the voltage profiles, distribution losses, costumer reliability, total operation costs etc. Those formulations were solved with several analytical and heuristic solution methods. Since the formulations comprise both the discrete (siting) and the continuous (sizing) variables, heuristic methods were more popular when compared to analytical solution methods. Both the traditional ones (Genetic algorithms (GA), harmony search (HS) method, differential evolution (DE) method) [3-5] and the new-generations (Cuckoo Search Algorithm - CSA), Grey Wolf Optimizer (GWO), etc.) [6, 7] were implemented. The following sections summarize the literature survey on optimal siting and sizing of renewable generation in distribution grid.

Several papers compared the performance of optimization algorithms in terms of achieved objective functions including the active power losses. For instance, in [8], the Tabu Search (TS) and Genetic Algorithm (GA) were compared for allocation of capacitor banks and distributed generators, and in [9], Harmony Search (HS) algorithm and Non-Dominated Sorting Genetic Algorithm II (NSGA-II) were compared. Modified Teaching-Learning-based Optimization Algorithm (MTLBO) was compared with GA and PSO again for minimizing the active power losses in [10].

An objective function representing the voltage profiles was improved in [11] where CSA was used to find optimal DG locations and their sizes in radial distribution networks. Similarly, [12] used the firefly algorithm (FA) for optimal sizing and siting of voltage controlled distributed generators. In [13], optimal DG siting and sizing was determined by comparing different scenarios using different numbers of DGs aiming to minimize power losses through the Whale Optimization Algorithm (WOA). Gray Wolf Optimizer (GWO) was used to minimize active power losses and voltage violations in [14]. The authors of [15] used GWO to minimize voltage deviations in distributions systems with PV penetrations. Also, mathematical and analytical approaches were used to solve similar problems [16,17].

This paper presents GWO-based optimal siting and sizing of PV units and WTs to improve the voltage profiles and so does to minimize the active power energy losses. The problem is adopted for Turkish regulations where renewable energy systems producing 1 MW or less power do not need a license, and are eligible for the specific Renewable Energy Support tariffs for the first ten years of their operation. Apart from

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available studies, all the summer, winter and spring/fall daily load patterns are taken into account to assess realistic values for this planning problem. The proposed technique is tested on 33-bus and 69-bus test systems with different scenarios.

The rest of the paper is organized as follows. Problem formulation is reviewed in Section II, solution technique GWO is briefly described in Section III. Section IV is devoted to the implementation of the proposed GWO algorithm and Section V summarizes the simulation results for test system applications. Finally, Section VI concludes the paper.

II. PROBLEM FORMULATION

There are several different formulations for voltage profile improvements. Among them, we preferred the bus voltage magnitudes in the feeder to be as close as each other. In this sense, optimization of the location and the sizes of PV and WTs to improve the voltage profiles in a distribution feeder is formulated as follows;

$$\underset{wrt\ \tilde{L}_R,\tilde{S}_R}{\textbf{minimize}} \sum_{i=1}^{N_T} \sum_{j=1}^{N_{BUS}} \sum_{k=j+1}^{N_{BUS}} \left[V_{ij} - V_{ik}\right]^2 \qquad (1)$$

$$\textbf{subject to} \begin{matrix} \tilde{f} = 0 \\ \tilde{g} \geq 0 \end{matrix}$$

where $\tilde{L}_R$ and $\tilde{S}_R$ denote the location and the size vectors of the renewables, respectively. N_T is the total optimization period in hours (72 hours in this study), N_{BUS} is the number of busses in the feeder, and V_{ij} is the bus voltage magnitude at bus j at hour-i. $\tilde{f}$ and $\tilde{g}$ vectors denote the equality and the inequality constraints, respectively. Equality constraints are load flow equations, where the inequality constraints represent the physical restrictions upon the dependent variables (line power flows, voltage magnitudes) and are generally held by using penalty factors. In this regard, optimization problem can be expressed as;

$$\underset{wrt\ \tilde{L}_R,\tilde{S}_R}{\textbf{minimize}} \sum_{i=1}^{N_T} \sum_{j=1}^{N_{BUS}} \left[\sum_{k=j+1}^{N_{BUS}} \left[V_{ij} - V_{ik}\right]^2 \right] + CP \qquad (2)$$

$$\textbf{subject to } \tilde{f} = 0$$

where P is the penalty factor defined below and C is the constant, balancing the penalty term and the main objective function.

$$P = \begin{cases} (v_{ij} - 0.95)^2 & if\ v_{ij} < 0.95 \\ 0 & if\ \ 0.95 \leq v_{ij} \leq 1.05 \\ (v_{ij} - 1.05)^2 & if\ \ v_{ij} > 1.05 \end{cases} \qquad (3)$$

Heuristic method based solution of the optimization equations given by (2) is as follows:

a. Generate initial control variables,
b. Solve load flow equations, $\tilde{f} = 0$, by using appropriate solution algorithm,
c. Evaluate the augmented objective function (OF),
d. Finalize the optimization cycle, if stopping criteria is achieved. If not, apply the operations of the heuristic algorithm and determine the off-spring control variables.
e. Go to case-b

In this study, load flow equations are solved with Forward/Backward Sweep Method. Forward/Backward Sweep method is generally used for radial network where Newton-Raphson Method is not effective solution algorithm due to high R/X ratio of the lines. The forward sweep is performed for node voltage calculations from the sending end to the far end of the feeder and laterals; whereas the backward sweep is used for branch current and/or power calculations from the far end to the sending end of the feeder and lateral. For more information on Forward/backward sweep algorithm one may refer to [18].

III. GREY WOLF OPTIMIZATION

Like other nature inspired algorithms GWO uses processes from nature. In GWO hunting behavior of grey wolves is modeled to solve optimization problems [19]. In grey wolf flocks, leaders are called alphas (α) and most of the important decisions are taken by them. There are also beta (β) and omega (ω) wolves whose duties are helping to take decisions of αs and obeying other types of wolves respectively. On the other hand there exists a fourth type: delta (δ), these are the wolves which do not comply with the previous three types of wolves.

Modeling the optimization process involves searching, encircling and attacking steps. Also the optimization process assumes the fittest solution as the α. Other good solutions are saved as well: second fittest and third fittest solutions are β and δ respectively. All other solution candidates are the ωs.

The first step encircling is modeled by using the equations specified in [25].

$$\vec{D} = \left|\vec{C}.\overrightarrow{X_p}(i) - \vec{X}(i)\right| \qquad (4)$$

$$\vec{X}(i+1) = \overrightarrow{X_p}(i) - \vec{A}.\vec{D} \qquad (5)$$

where i indicates the current iteration, $\vec{A}$ and $\vec{C}$ are coefficient vectors, $\overrightarrow{X_p}$ is the position vector of the prey, and $\vec{X}$ indicates the position vector of a grey wolf.

The vectors $\vec{A}$ and $\vec{C}$ are calculated as follows:

$$\vec{A} = 2\vec{a}.\overrightarrow{r_1} - \vec{a} \qquad (6)$$

$$\vec{C} = 2.\overrightarrow{r_2} \qquad (7)$$

where components of $\overrightarrow{r_1}, \overrightarrow{r_2}$ are random vectors in [0, 1] range and the values $\vec{a}$ linearly decrease from 2 to 0 during the optimization process.

Different wolf types are expressed as follows [25]. These situations are expressed in the following expressions:

$$\overrightarrow{D_\alpha} = \left|\overrightarrow{C_1}.\overrightarrow{X_\alpha}(i) - \vec{X}(i)\right| \qquad (8)$$

$$\overrightarrow{D_\beta} = \left|\overrightarrow{C_1}.\overrightarrow{X_\beta}(i) - \vec{X}(i)\right| \qquad (9)$$

$$\overrightarrow{D_\delta} = \left|\overrightarrow{C_1}.\overrightarrow{X_\delta}(i) - \vec{X}(i)\right| \qquad (10)$$

These equations determine the distances and new locations are calculated by using following equations [25].

$$\overrightarrow{X_1} = \overrightarrow{X_\alpha}(i) - \overrightarrow{A_1}.\overrightarrow{D_\alpha} \qquad (10)$$

$$\overrightarrow{X_2} = \overrightarrow{X_\beta}(i) - \overrightarrow{A_2}.\overrightarrow{D_\beta} \qquad (11)$$

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$$\overrightarrow{X_3} = \overrightarrow{X_\delta}(i) - \overrightarrow{A_3}.\overrightarrow{D_\delta} \quad (12)$$

Position update process for ω values are [25]:

$$\vec{X}(i+1) = \frac{\overrightarrow{X_1}+\overrightarrow{X_2}+\overrightarrow{X_3}}{3} \quad (13)$$

IV. Implementation of The Proposed GWO Algorithm

The steps of GWO implementation to optimize the locations and the sizes of DG units.

Step-1. Input test system data, number of PVs and WTs, location and size restrictions, maximum number of iterations, number of search agents and the objective function.

Step-2. Initialize a search agent population (Initial locations and sizes) randomly in predefined solution space; solve power flow equations and calculate the OF values for each solution candidate and determine α, β and δ parameters.

Step-3. Check the positions (locations and sizes) constraints of each search agent to determine the quality of the solution; calculate the OF. Note that, during the solution process if there are any solution candidates in the same locations those solution candidates are neglected.

Step-4. The best positions of the α, β and δ are updated assuming that they have better knowledge about the potential location of optimum; whereas ω is obliged to update its positions according to the position of the best search agent.

Step-5. The new positions of the search agents are determined and the whole process is repeated.

Step-6. The stopping condition is set as maximum iterations. When the condition is satisfied, then the simulation will be stopped and the optimum location and size of multi-DG units which satisfy all the specified constraints of the distribution system will be obtained.

This procedure is described through the flowchart presented in Fig.1.

V. Test Systems and Simulations

A. Test Systems

The proposed formulations are applied to 33-bus and 69-bus radial distribution test systems. Line and load data of the systems are given in [20-21]. For both test systems, it is assumed that all the buses can be taken as candidate for DG units' placement except the one connected to the main supply. We also assume that 2 or more units cannot be connected to the same bus.

Scaled daily load curve is shown in Fig.2. for the three consecutive days (72 hours), representing 3 seasonal days; namely, the first 24 hours correspond to average of spring and fall loads, the second 24 hours correspond to winter load and finally the last 24 hours show the summer loads. The load data is given in [22].

B. PV and WIND Parks

DGs are considered to be either PV or Wind Park units. Due to governmental incentives provided for small scale DGs in Turkey, PV and Wind Park unit outputs are limited with 1 MW. Scaled PV and Wind Park outputs, taken from [23-24] are shown in Fig.2, together with daily load curves.

Without loss of generality, five different scenarios, each representing the availability of certain number of DGs are considered. Fixed penalty coefficient, C =100 is applied in simulations where the number of search agents are assumed to be 30.

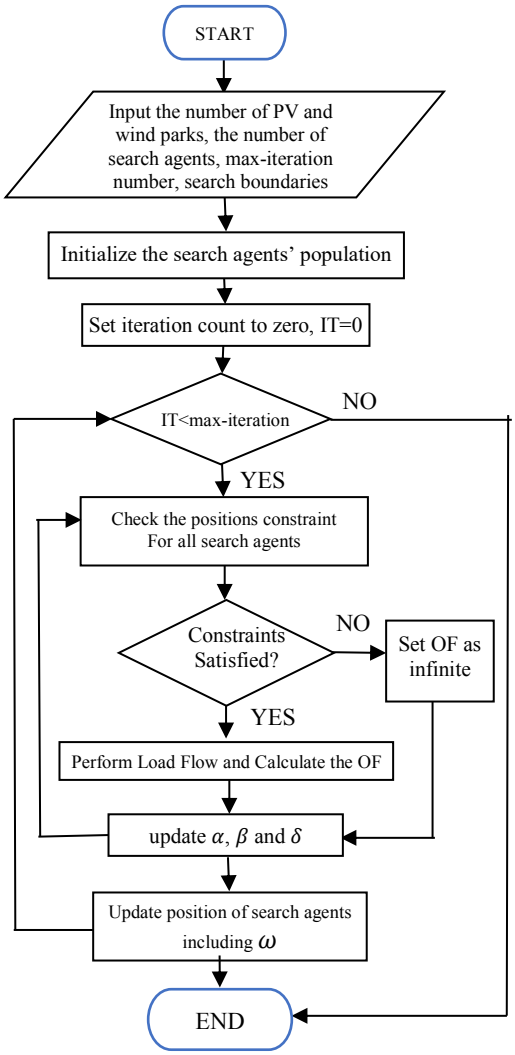


Fig.1. The flow chart of the proposed GWO-based solution algorithm

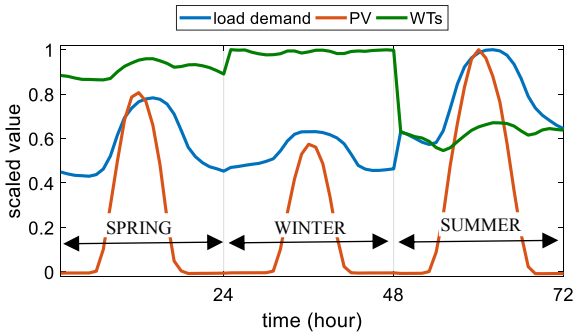


Fig.2. Scaled load curve, and DG outputs for three seasons.

C. Simulation Results

The maximum active and reactive loads of 33 bus test system are 3.715 MW and 2.3 MVAR at 3rd day (Summer), hour 14, respectively. Base case (without DG) real and reactive power losses are 201.9 kW and 134.6 kVAR at that time period, respectively. Under these conditions, minimum voltage magnitude is 0.913 pu at bus 18 and the OF value is

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11.9308. Optimal DG locations and sizes obtained by using GWO are illustrated in Table I, for the five scenarios.

TABLE I. OPTIMAL DG SITING AND SIZINGS FOR 33 BUS TEST SYSTEM

Case#	*OF*	*Maximum Iteration*	*PV/WTs*	*PV*		*WTs*	
				site	*size [kW]*	*site*	*size [kW]*
BASE	*11.930*	-	-	-	-	-	-
1	0.2399	10000	1/1	15	1000	29	1000
2	0.1653	20000	2/1	33	1000	8	1000
				18	566		
3	0.1528	20000	2/2	15	644	17	238
				33	947	32	326
4	0.1528	20000	3/2	15	644	17	238
				33	947		
				22	0	32	326
5	0.1524	30000	3/3	8	354	17	239
				16	502	32	332
				33	841	2	0

The minimum OF is found to be 0.1524 for Case-5, where 3 PV and 2 WTs are installed. However, the total installed DG capacity (probably the cost) is minimum for Case-3 and Case-4 (2155 kW); where case 5 involves 2268 kW of DG installation. Beyond this point, increasing the number of DGs does not provide better objective function values, so doesn't provide any more voltage improvement.

The voltage profiles of each bus along several hours are shown in Fig.3 for the base case and for Case-5 scenarios. Each curve shows the voltage magnitude of the busses at a certain hour. Note that only limited number of hours are depicted in the figure for the sake of clear visibility. It is clear that there are so many busses whose voltage magnitudes are less than the lower voltage limits (0.95 p.u.) for the base case conditions. It can also be recognized that under-voltage conditions are more severe for peak loading hours (e.g. summer 12th hour, summer 18th hour). When 3 PV and 2 WTs is connected to the system, there are only small violations of the lower voltage limit at busses 14-18 and 31-33 at Summer 6 and 7 pm.

It can be understood that Bus-18 is the most critical one, showing the highest under voltage cases for base-case conditions. Fig.4 shows the Bus-18 voltage magnitudes for the base case and for the other five cases. One can easily realize that all cases provide improvements in Bus-18 voltage magnitude with respect to base case values. Among them, Case-5 is the best one, eliminating almost all under voltage problems besides a few during summer peaks.

Power losses for each hour and the total energy losses along 72 hours are also calculated. They are illustrated in Table II, where the relative losses show the percentage of the losses with respect to corresponding loads. Note that the range of relative hourly losses (RHL), their maximum value and energy losses are decreased in case of DG penetrations. The results show that, all the 5 cases provide an energy saving of 46.8-49.1 % energy saving with respect to base case operating conditions. Moreover, all cases provide significant decreases in terms of hourly losses.

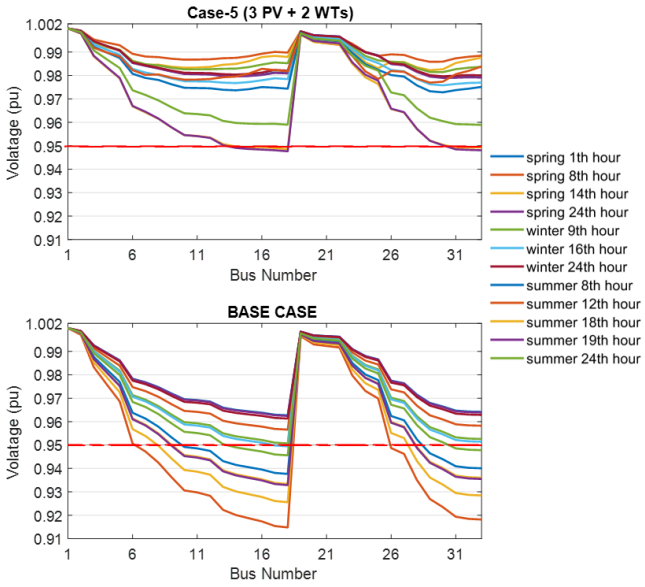


Fig.3. 33-Bus Test System Voltage profiles for 72 hours.

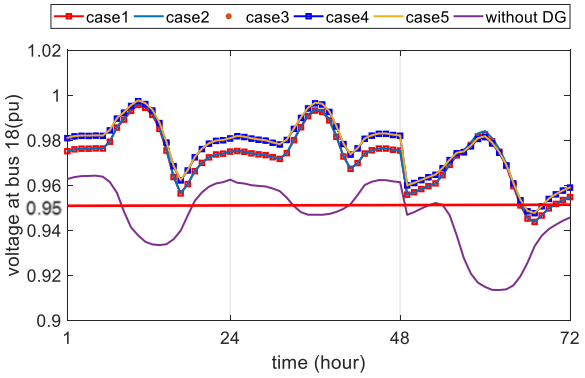


Fig.4 Voltage profile for bus number 18 for 3 days simulation

TABLE II. POWER AND ENERGY LOSSES FOR 33 BUS TEST SYSTEM

Case#	RHL [%]		Energy losses along 72 hours	
	minimum	*maximum*	*MWh*	*Relative[%]*
BASE	2.1593	5.4345	5.899	3.5070
1	1.2248	2.7963	3.139	1.8664
2	1.1313	2.8940	3.002	1.7851
3	1.1360	3.0513	3.130	1.8613
4	1.1360	3.0513	3.130	1.8613
5	1.1230	3.0446	3.062	1.8207

69-bus test radial distribution system has the total load of 3.80 MW and 2.69 MVAr for the 3rd day (Summer) at hour 14. Base case (without DG) real and reactive power losses are 224.5 kW and 101.9 kVAR at that time period, respectively. Under these conditions, minimum voltage magnitude is 0.910 pu at bus 65 and the OF value is 8.9876 for the base case conditions, where there is no any installed DGs. Optimal DG locations and sizes obtained by using GWO are illustrated in Table III, for the five scenarios.

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TABLE III. OPTIMAL DG SITING AND SIZINGS FOR 69 BUS TEST SYSTEM

Case#	OF	Maximum Iteration	PV/WTs	PV		WTs	
				site	size [kW]	site	size [kW]
BASE	8.9876	-	-	-	-	-	-
1	0.2105	10000	1/1	65	1000	63	1000
2	0.1541	20000	2/1	65	1000	63	629
				12	868		
3	0.1291	20000	2/2	65	1000	63	629
				69	854	31	0
4	0.1256	30000	3/2	61	214	63	612
				15	405		
				65	1000	4	13
5	0.1256	30000	3/3	65	1000	63	612
				61	214	4	13
				15	405	2	0

The best OF value is found to be 0.1256 for Case-4 and Case-5, where 3 PV and 2 WP are installed. Note that the total installed DG capacity is also minimum (2244 kW) for those optimal cases. Increasing the number of DGs beyond that point does not provide better objective function values, so doesn't provide any more voltage improvement.

Some of the representative voltage profiles of each bus along 72 hours are shown in Fig.5 for the base case and for Case-5 scenarios. Each curve shows the voltage magnitude of the busses at a certain hour. It is clear that there are so many busses whose voltage magnitudes are less than the lower voltage limits (0.95 p.u.) for the base case conditions. It can also be recognized that under-voltage conditions are more severe for peak loading hours (e.g. Summer 2.00 p.m). When 3 PV and 2 WTs is connected to the system, there are only small violations of the lower voltage limit at busses 61-65 at Summer 6,7 pm.

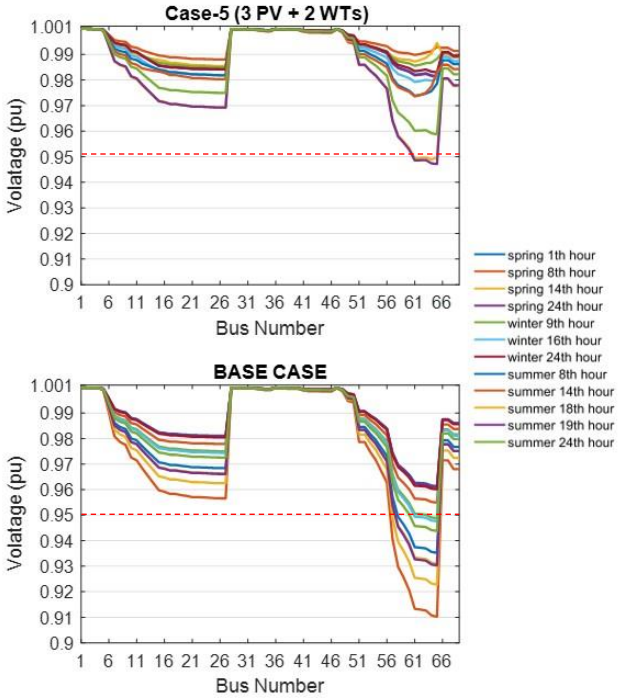


Fig.5 69-Bus Test System Voltage profiles for 72 hours.

Bus-65 is the most critical one, showing the highest under-voltage cases for base-case conditions. Fig.6 shows the Bus-65 voltage magnitudes for the base case and for the five cases. One can easily realize that all cases provide improvements in Bus-65 voltage magnitude with respect to base case. Among them, Case-4 is the best one, eliminating almost all under-voltage problems besides a few during summer peaks.

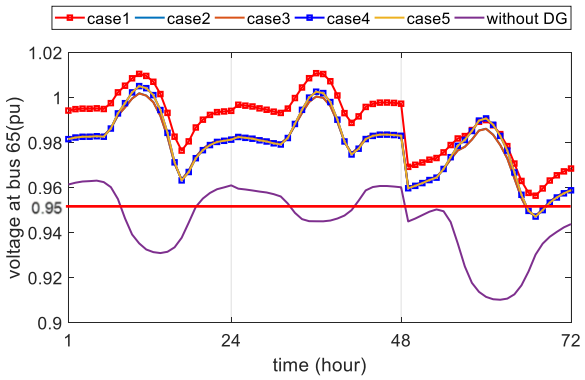


Fig.6 Voltage profile for bus number 65 for 3 days simulation

Power losses for each hour and the total energy losses along 72 hours are illustrated in Table IV. Note that the range of relative hourly losses (RHL), their maximum value and 72-hour energy losses are decreased in case of DG penetrations. Among them, Case-3, Case-4 and Case-5 seem to provide similar benefits. The results show that, all the 5 cases provide an energy saving of 53.0-55.0 % energy saving with respect to base case operating conditions. Moreover, all cases provide significant decreases in terms of hourly losses.

TABLE IV. POWER AND ENERGY LOSSES FOR 69 BUS TEST SYSTEM

Case#	RHL [%]		Energy losses along 72 hours	
	minimum	maximum	MWh	Relative[%]
BASE	2.31	5.91	6.5136	3.7832
1	0.96	2.62	2.9280	1.7006
2	1.01	3.07	3.0157	1.7516
3	1.02	3.07	3.0486	1.7706
4	1.03	3.10	3.0124	1.7496
5	1.03	3.10	3.0124	1.7496

VI. CONCLUSION

This paper has presented optimal sitting and sizing of PV and WP units to improve the voltage profiles and so does to minimize the active power energy losses in radial distribution networks. The problem is adopted for Turkish regulations where renewable energy systems producing 1 MW or less power do not need a license, and are eligible for the specific Renewable Energy Support tariffs for the first ten years of their operation.

All the summer, winter and spring/fall daily load patterns are taken into account to assess realistic values for this planning problem. The proposed technique is tested on 33-bus and 69-bus test systems with different number of renewable penetration levels.

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The results show that almost all operating cases provide improvements in voltage magnitude profiles. Among them, some configurations provide up to 50% of energy savings in losses. At the second phase of the study, objective function will be extended to include the costs of renewable generation and more realistic evaluation will be provided.

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